

Syllabus :- MATRICES

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- If $B = B^2$ and $I - B = A$ then (A, B are square matrices of same order)
 - $A^2 = B$
 - $A^2 = A$
 - $A^2 = I$
 - $A^2 = -A$
- For what values of n ($n \in \mathbb{N}$), $A^n = nA - nI + I$ where $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and I is identity matrix ?
 - $n = 1$ only
 - $n = 2$ only
 - all values of n
 - no value of n
- Let A be a 2×2 matrix with real entries. Let I be the 2×2 identity matrix and $\text{tr}(A)$ denotes sum of diagonal entries of A . Assume that $A^2 = I$. Then which of the following statements is true
 - $A \neq I$ and $A \neq -I \Rightarrow |A| = -1$
 - $A \neq I$ and $A \neq -I \Rightarrow \text{tr}(A) \neq 0$
 - $\text{tr}(A) = 0$
 - $|A| = 1$
- How many of the following statements are true ?

S_1 : If B is symmetric matrix then ABA^T is symmetric

S_2 : If A^4 is singular matrix then A is also singular

S_3 : If $AB = O$ and $|A|$ is non zero then B must be a null matrix

S_4 : If $|A| \neq 0$ and $(\text{adj } A)B \neq 0$ then matrix equation $AX = B$ has no solution

 - 1
 - 2
 - 3
 - 4
- Let A be a 3×3 matrix such that $A^2 - 5A + 7I = O$.

Statement – I : $A^{-1} = \frac{1}{7}(5I - A)$.

Statement – II : The polynomial $A^3 - 2A^2 - 3A + I$ can be reduced to $5(A - 4I)$. Then

 - Statement-I is false, but Statement-II is true.
 - Both the statements are false.
 - Both the statements are true.
 - Statement-I is true, but Statement-II is false.
- For two 3×3 matrices A and B , let $A+B = 2B'$ and $3A+2B = I_3$, where B' is the transpose of B and I_3 is 3×3 identity matrix. Then :
 - $10A+5B = 3I_3$
 - $5A+10B = 2I_3$
 - $3A+6B = 2I_3$
 - $B + 2A = I_3$
- Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^{20}$. Then the sum of the elements of the first column of B is :
 - 210
 - 211
 - 251
 - 231

8. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a 2×2 real matrix. If $A - \alpha I$ is invertible for every real number α , then
- (1) $bc > 0$ (2) $bc = 0$ (3) $bc < \min\left(0, \frac{ad}{2}\right)$ (4) $a = 0$
9. Let S be the set of all real matrices, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that $a + d = 2$ and $A^T = A^2 - 2A$. Then S
- (1) has exactly two elements (2) has exactly four elements
(3) is an empty set (4) has exactly one element
10. If $S_k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$, $k \in \mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers, then $(S_2)^n (S_k)^{-1}$, for $n \in \mathbb{N}$, is :
- (1) S_{2n-k} (2) S_{2n+k-1} (3) S_{2n+k} (4) S_{2n+k-1}
11. If A is a non singular matrix satisfying $A = AB - BA$, then which one of the following holds true
- (1) $\det B = 0$ (2) $B = 0$ (3) $\det A = 1$ (4) $\det(B+I) = \det(B-I)$
12. If ω is a imaginary cube root of unity and $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix}$, then A^{-1} is
- (1) $\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega^2 & 1 & \omega \\ \omega & \omega^2 & 1 \end{bmatrix}$ (2) $\frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \\ 1 & 1 & 1 \end{bmatrix}$ (4) $\frac{1}{2} \begin{bmatrix} 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \\ 1 & 1 & 1 \end{bmatrix}$
13. The homogeneous system of equations
- $$\begin{bmatrix} 2 & a+b+c+d & ab+cd \\ a+b+c+d & 2(a+b)(c+d) & ab(c+d)+cd(a+b) \\ ab+cd & ab(c+d)+cd(a+b) & 2abcd \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$
- has non-trivial solutions only if
- (1) $a + b + c + d = 0$ (2) $ab + cd = 0$
(3) $ab(c+d) + cd(a+b) = 0$ (4) for any a, b, c, d
14. If $|A - B| \neq 0$, $A^4 = B^4$, $C^3A = C^3B$, $B^3A = A^3B$, then matrix $A^3 + B^3 + C^3$ is
- (1) null matrix (2) non-zero matrix (3) non-singular (4) unit matrix
15. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ then trace of matrix $\text{adj}(\text{adj}A)$ is
- (1) 1 (2) 2 (3) 3 (4) 4
16. If $f(x) = \begin{bmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\alpha) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{bmatrix}$ and $|f(0)| = -2$ then $\sum_{r=0}^{30} |f(r)|$ equals
- (1) 2 (2) 30 (3) -62 (4) 62
17. If A and B are square matrices of order 3 such that $|A| = -2$, $|B| = 1$ then $|A^{-1} \text{adj} B^{-1} \text{adj}(2A^{-1})| =$
- (1) 8 (2) -8 (3) 1 (4) -2
18. If M is a square matrix of order 2 then $-\text{tr}(M^2) + (\text{tr}M)^2$ is ($\text{tr}(P)$ denotes trace of matrix P)
- (1) $\frac{|M|}{2}$ (2) $2|M|$ (3) $3|M|$ (4) $4|M|$

19. If $A = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$, then $A + 2A^T$ equals -
 (1) A (2) $-A^T$ (3) A^T (4) $2A^2$
20. The product of the matrices $A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$ is a null matrix
 if $\theta - \phi =$
 (1) $(2n+1)\frac{\pi}{2}$ (2) $n\pi$ (3) $2n\pi$ (4) $\frac{n\pi}{2}$

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. How many of the following statements are true for non-singular square matrices A and B of order 3 ?
 S_1 : If $AB = BA$ then $A^4B = BA^4$ and $(AB)^3 = A^3B^3$
 S_2 : $||A|A| = |A|^2$
 S_3 : Trace of $AB =$ trace of BA
 S_4 : $(A + B)^{-1} = A^{-1} + B^{-1}$
22. Let $A = [a_{ij}]$ be a 3×3 matrix whose determinant is 5. Then the determinant of the matrix $B = [2^{i-j} a_{ij}]$ is
23. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix}$. If $10A^{10} + \text{adj}(A^{10}) = B$, then $\frac{30}{80} (b_1 + b_2 + b_3 + b_4)$ is equal to
24. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$ and $A^{-1} = \frac{1}{6} (A^2 + cA + dI)$ where I is identity matrix of order 3 then $c + d =$
25. If A is a n rowed square matrix, $A = [a_{ij}]$ where $a_{ij} = \left[\frac{i}{j} \right]$, [] denotes greatest integer, then $\det(A) =$

Answer Key_DPP-06

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|-----|-----|-----|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1. | (2) | 2. | (3) | 3. | (1) | 4. | (3) | 5. | (3) | 6. | (1) | 7. | (4) |
| 8. | (3) | 9. | (3) | 10. | (1) | 11. | (4) | 12. | (2) | 13. | (4) | 14. | (1) |
| 15. | (1) | 16. | (3) | 17. | (2) | 18. | (2) | 19. | (3) | 20. | (1) | 21. | (2) |
| 22. | (5) | 23. | (42) | 24. | (5) | 25. | (1) | | | | | | |